Blue screen (BSOD) troubleshooting

Article KB0010049 | Owner: Desktop Support | Subcategory: Blue screen / crash | Applies to: Dell Latitude 5440, Dell Latitude 7440, Lenovo ThinkPad T14, Lenovo ThinkPad X1 Carbon, HP EliteBook 840, Windows 11 23H2/24H2

Summary

This article guides users and service-desk agents through diagnosing and resolving blue-screen (BSOD) crashes on Acme Global Windows laptops. It outlines common symptoms, typical causes, a self-help troubleshooting flow, and the detailed steps Desktop Support follows to isolate hardware, driver, or update issues. Follow the verification steps to confirm stability before closing the ticket.

Symptoms

- STOP error 0x0000007E - SYSTEM_THREAD_EXCEPTION_NOT_HANDLED
- STOP error 0x0000001A - MEMORY_MANAGEMENT
- STOP error 0x000000D1 - DRIVER_IRQL_NOT_LESS_OR_EQUAL
- STOP error 0x00000050 - PAGE_FAULT_IN_NONPAGED_AREA
- Blue screen with message: "Your PC ran into a problem and needs to restart. We're collecting some error info, and then we'll restart for you."
- System restarts unexpectedly after Windows Update or driver install
- Frequent crashes during graphics-intensive tasks (video playback, gaming, CAD)

Common causes

- Faulty or outdated graphics driver (NVIDIA/AMD/Intel)
- Recent Windows quality or feature update that introduced a bug
- Corrupted system files or registry
- Defective RAM or memory module
- Overheating due to blocked vents or failing fan
- Incompatible third-party software or antivirus
- BitLocker recovery key mismatch after BIOS/firmware change

Overview

A blue screen (BSOD) indicates a critical kernel error that forces Windows to halt. The stop code displayed on the screen points to the faulty component-most often a driver, memory, or recent update. At Acme Global, the majority of BSOD tickets are resolved by rolling back a problematic graphics driver, replacing bad RAM, or uninstalling a recent Windows update. This article provides a structured approach for end-users to perform basic checks and for Desktop Support agents to conduct deeper diagnostics.

Self-Help Troubleshooting Steps

Follow these steps if you experience a BSOD. They are safe to perform on your own and can resolve many common issues without needing to contact the service desk. If the problem persists after completing the procedure, gather the stop code and any dump file information and open a ticket via the IT self-service portal, phone extension 4357, or the Teams "IT Help" channel.

1. 1. Note the exact stop code and any filename shown on the blue screen (e.g., "STOP 0x0000007E, nvlddmkm.sys"). Take a photo or write it down before the system restarts.
2. 2. Boot into Safe Mode: hold the Shift key while clicking Restart, then choose Troubleshoot -> Advanced options -> Startup Settings -> Restart -> press 4 for Safe Mode.
3. 3. In Safe Mode, open Settings -> Windows Update -> View update history -> Uninstall updates. Remove the most recent quality or feature update (look for the date that matches when the crashes began).
4. 4. Open Device Manager (right-click Start -> Device Manager). Expand Display adapters, right-click your graphics adapter, select Properties -> Driver -> Roll Back Driver if the option is available. If Roll Back is greyed out, choose Uninstall device, check "Delete the driver software for this device", then reboot.
5. 5. Run Windows Memory Diagnostic: type "mdsched.exe" in the Start menu, select Restart now and check for problems. The system will reboot and test RAM; note any errors reported after the restart.

6. 6. After each step, test normal use for at least 30?minutes. If the BSOD does not return, the issue is likely resolved.

Verification and Closure

After applying any fix, it is essential to confirm that the system remains stable before closing the ticket. Ask the user to resume normal work activities for a defined observation period and verify that no further BSOD events occur. Collect final logs and update the ticket with the resolution details.

1. 1. Have the user run their typical workload (e.g., email, Teams, line-of-business applications) for at least 2?hours. If they use graphics-intensive applications, include a 15-minute stress test (e.g., video rendering or CAD).
2. 2. Check the Reliability Monitor (Control Panel -> Security and Maintenance -> Reliability Monitor) for any new critical errors or warnings.
3. 3. If no new critical events appear, close the ticket with resolution description (e.g., "Rolled back graphics driver version 31.0.15.4625 to 30.0.14.1188; no further BSOD observed.") and add the stop code and any relevant file names to the resolution notes.
4. 4. Advise the user to report any recurrence immediately; if a second BSOD occurs, re-open the ticket and escalate to the Senior Desktop Support lead.

When to escalate

If the BSOD persists after completing the Service Desk Procedure (steps 1-5) or if the analysis reveals a hardware fault beyond RAM (e.g., motherboard, CPU), escalate to the Senior Desktop Support Lead or the Hardware Engineering team. Include in the escalation: the stop code, offending module name (from WinDbg), results of memory and driver tests, a list of troubleshooting steps already performed, and any relevant minidump files attached to the ticket.

Keywords: BSOD, blue screen, stop error, memory dump, graphics driver, Windows update, RAM test, Safe Mode, Device Manager, sfc, DISM, WinDbg, Reliability Monitor, Intune, Company Portal, overheating, fan